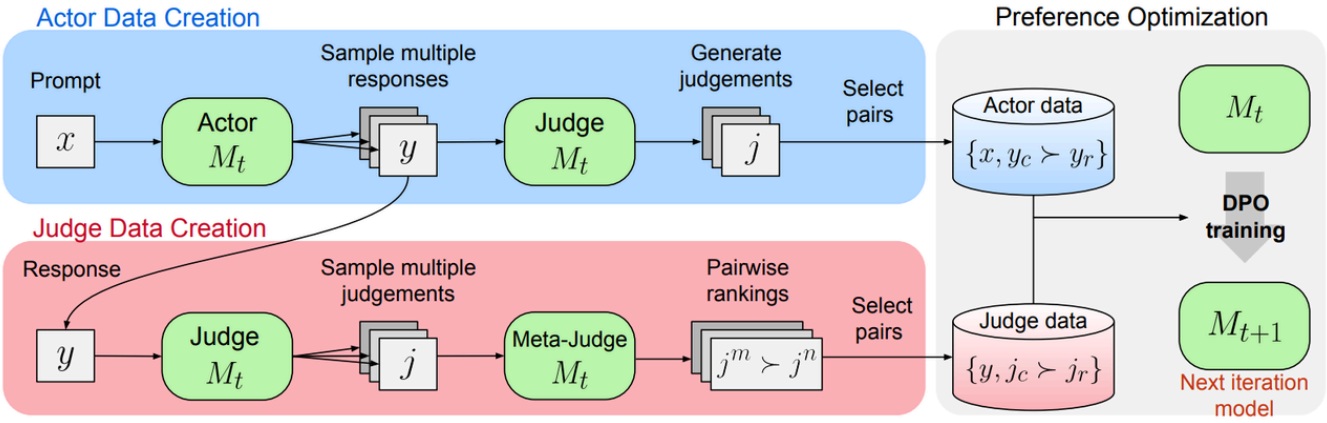
META-REWARDING LANGUAGE MODELS: Self-Improving Alignment with LLM-as-a-Meta-Judge

> #DPO #RewardModel



- What problems does this framework solve?
  - Normally, the inference ability of LLMs is improving. However, there is not a way to improve both its ability to be a judge to run self-judge and its generation's ability simultaneously.
  - Hence, the framework of meta-rewarding language model is proposed to improve both without the labeling from humans.
- What does this framework do?
  - This framework is DPO-based finetuning with two sets of data prepared: the dataset of actor and the dataset of judge (The same model will play three roles: actor, judge, and meta-judge).
  - Actor Dataset
    - For each input, we sample a set of answers of K. For each of the answers, let N judges give it a score. Average the score as its final score.
    - Select a ρ as a control param. Among the K answers, we select the shortest answer from the answers with a score in $[(1-\rho)S_{max} + \rho S_{min}, S_{max}]$ as the +answer of the actor data and longest answer with a score in $[S_{min}, (1-\rho)S_{min} + \rho S_{max}]$ as the -answer.
    - This can teach the model not to generate long-but-useless answers.
  - Judge Dataset
    - We select the answer with the highest variance in its N scores as the start, as this means that the judge is not very sure about its decision.
    - Let the meta-judge evaluate the N ratings pair-wisely with CoT based prompts. The order will be swapped to avoid the positional preference.
    - Use Elo rating to further evaluate each pair.

      $\arg\max_{\epsilon} \sum_{m,n} B_{mn} \log \frac{1}{1+e^{\epsilon_m - \epsilon_n}}$
    - Select the judge with the highest Elo score and the lowest Elo score as the +- pair of the judge dataset.

    - Use these two datasets, run DPO.
- Experiment Results
    - Instruction Following Capability Surge: On the AlpacaEval 2 dataset, its length control (LC) win rate surged dramatically from 22.9% to 39.4%, even surpassing the powerful GPT-4-0314. On the complex problem-assessing Arena-Hard dataset, its score also improved from 20.6% to 29.1%.
    - Significantly Enhanced Judgment Capabilities: Experiments demonstrate that incorporating a meta-judge not only effectively controls response length but also substantially elevates the model's performance as a “referee.” Compared to self-generated baseline models, the trained judge exhibits greater consistency in scoring with both the powerful GPT-4 and human preferences.
    - Surpassing Strong Baselines: This fully self-evolutionary approach, requiring no additional human supervision, outperforms strong baseline models like SPPO—which heavily rely on external human and GPT-4 training data.
- Work Cited
    - Paper
    - NotebookLM
    - Chat History